
Wine Score Prediction with K-Nearest Neighbors and Random Forest

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Abstract

We study a supervised regression problem on a real-world consumer dataset: predicting expert wine scores from basic wine metadata. The target variable is the critic's score ("points"), and the input features include price and several categorical descriptors such as country, province, grape variety, and winery.

This project extends my mid-term work in three directions. First, I introduce a simple mean baseline to provide a clear reference point. Second, I replace a single train-test split with k-fold cross-validation to obtain more reliable estimates of generalisation error. Third, I complement the original k-nearest neighbours (KNN) regressor with a non-linear tree-based model (Random Forest) and compare them under a unified preprocessing pipeline.

On a random subset of 10,000 wines evaluated with 3-fold cross-validation, both KNN and Random Forest substantially outperform the mean baseline. The baseline yields an MAE of about 2.48 and an RMSE of 3.02 with $R^2 \approx 0.00$, while KNN reduces the MAE to 1.91 and raises R^2 to 0.36. The Random Forest achieves the best performance, with an MAE of 1.84, an RMSE of 2.32, and R^2 of 0.41. Feature importance analysis shows that price is the dominant predictor, followed by winery, variety, and region. These results illustrate how adding baselines, cross-validation, and a stronger non-linear model improves both the credibility and the insightfulness of the analysis.

1 Datasets

The dataset used in this project is the WineEnthusiast wine review dataset (winemag-data-130k-v2), which contains professional reviews of wines from multiple countries and regions. Each instance corresponds to a single wine and includes a textual description, basic structured attributes, the retail price, and an expert-assigned score (points).

For this project, I focus on a subset of structured features that are suitable for classical machine learning models:

- Target variable

- points: integer score assigned by a professional wine critic.

- Input features

- Numerical: price (USD).

- Categorical: country, province, variety, winery.

Rows with missing points or price are removed. The remaining missing values in categorical features are handled by imputation in the preprocessing pipeline (Section 2.1).

Table 1 summarises the basic characteristics of the filtered dataset used in this project.

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Table 1: Basic characteristics of the wine review dataset (after filtering).

Dataset	Task	# Instances	# Attributes	Target range (points)	Mean $\pm$ Std (points)	Median price (USD)
Wine reviews	Regression	120,975	5	80–100	88.42 $\pm$ 3.04	25.0

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1.1 Data characteristics

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Figure 1 shows the distribution of wine scores. Most wines fall between 80 and 95 points, with a slight skew towards higher scores. Because this range is narrow, prediction errors of 2–3 points are already meaningful.

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Wine prices are much more heterogeneous. The raw distribution is highly right-skewed, with many inexpensive wines and a long tail of very expensive bottles. Figure 2 shows the distribution of the log-transformed price $\log_{10}(\text{price})$, which is closer to normal and more suitable for modelling. Figure 3 plots $\log_{10}(\text{price})$ against points on a 3,000-sample subset. There is a clear positive trend—more expensive wines tend to receive higher scores—yet substantial scatter, suggesting that price alone cannot fully explain the variation and that categorical attributes such as region, variety, and winery may provide additional signal.

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Figure 1: Histogram of wine scores (points).

Figure 2: Histogram of log-transformed price ($\log_{10}(\text{price})$).Figure 3: Scatter plot of $\log_{10}(\text{price})$ vs points on a 3,000-sample subset.

1.2 Why is this an interesting dataset?

This dataset is interesting for several reasons. First, wine ratings are economically relevant: they influence consumer choices, pricing strategies, and inventory decisions. Predicting scores from basic features could support recommendation or pricing tools for retailers and producers. Second, the data combine numerical and high-cardinality categorical features (thousands of wineries and hundreds of varieties and regions), which creates a realistic challenge for preprocessing and model selection. Third, the relationships between price, origin, variety, and score are highly non-linear and involve interactions, making the dataset suitable for comparing different non-linear models such as KNN and tree ensembles.

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2 Methods

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This section describes the preprocessing pipeline and the three models used in the experiments. The design explicitly incorporates the mid-term feedback: we add a baseline model, implement cross-validation, and compare two non-linear models under the same preprocessing.

2.1 Preprocessing and basic feature engineering

To ensure consistent treatment of features across models and folds, I implement the following preprocessing steps using scikit-learn’s ColumnTransformer and Pipeline:

•Missing values.

Rows with missing target (points) or missing price are removed.

Remaining missing values in categorical features (country, province, variety, winery) are imputed with the most frequent category.

•Numerical feature (price).

After imputation, I apply the $\log_{10}$ transformation to price to reduce the impact of extreme outliers and stabilise variance.

The transformed price is then standardised using StandardScaler. Standardisation is essential for distance-based models like KNN; although Random Forests are scale-invariant, sharing the same preprocessing pipeline across models keeps the comparison fair.

•Categorical features.

91 All categorical attributes are one-hot encoded using OneHotEncoder with handle_unknown="ignore".
92 This results in a high-dimensional sparse representation, but tree-based ensembles and KNN can still
93 handle it, especially on the subsampled dataset.

94 I do not manually create polynomial or interaction features, since Random Forests can automatically
95 capture non-linear relationships and high-order interactions between attributes through their hierarchical
96 splits.

97 98 **2.2 Models**

99 I consider three models of increasing complexity:

100 **2.2.1 Mean baseline (DummyRegressor)**

101 The baseline model simply predicts the mean score of the training set for every instance. It completely
102 ignores the input features and serves as a lower bound. Any useful model should significantly
103 outperform this baseline. Including such a trivial model directly responds to the mid-term comment
104 about adding a baseline to improve the credibility of performance claims.

105 **2.2.2 K-nearest neighbours regressor**

106 The KNN regressor predicts the score of a wine as the average score of its k nearest neighbours in the
107 transformed feature space (after preprocessing). Distance is measured using the Euclidean metric over
108 the standardised numerical feature and the one-hot-encoded categorical features.

109 KNN is a natural choice for this dataset:

- 110 • Wines with similar price, country, region, variety, and winery are expected to have similar
111 quality and therefore similar scores.
- 112 • KNN is non-parametric and can flexibly adapt to local structure without assuming a specific
113 functional form.

114 In this project, I use k = 5 neighbours with uniform weighting. This choice provides a good bias–variance
115 trade-off on preliminary experiments, and it is consistent with common practice for medium-sized
116 tabular datasets.

117 **2.2.3 Random Forest regressor**

118 The Random Forest regressor is an ensemble of decision trees trained on bootstrapped samples with
119 random feature selection at each split. It can model complex non-linear relationships and interactions
120 between features and often performs well on high-dimensional tabular data:

- 121 • It can capture interactions such as “price × region” or “variety × winery” without explicit
122 feature engineering.
- 123 • It is robust to outliers and can handle mixed-type inputs after one-hot encoding.
- 124 • It provides feature importance measures that help interpret what the model has learned.

125 To control complexity and reduce the risk of overfitting, I use a moderately sized Random Forest with
126 a limited maximum depth and a fixed number of trees, as summarised in Table 2.

127 **2.3 Hyperparameters and model complexity**

128 The key hyperparameters used for each model are shown in Table 2.

129 *Table 2: Key hyperparameters of the models.*

Model	Key hyperparameters
Baseline	strategy = "mean"
KNN	n_neighbors = 5, weights = "uniform"
Random Forest	n_estimators = 50, max_depth = 12, random_state = 42, n_jobs = -1

130 For the Random Forest, limiting max_depth and using a moderate number of trees acts as a form of pre-
131 pruning, which reduces variance.

132 133 **3 Experimental Setup**

134 **3.1 Subsampling and cross-validation**

135 The full filtered dataset contains over 120,000 instances. To keep the experiments computationally
136 manageable, I randomly sample 10,000 wines from this dataset and use the same subset for all model
137 comparisons.

138 All models are evaluated using 3-fold cross-validation with shuffled splits and a fixed random seed

(random_state = 42) to ensure reproducibility. Each fold uses two-thirds of the data for training and one-third for testing.

Using cross-validation instead of a single train–test split has several advantages:

- It reduces the dependence of the results on a particular random split.
- It provides a more reliable estimate of the generalisation error.
- It is consistent with best practices emphasised in the course and directly addresses the mid-term feedback regarding the evaluation procedure.

3.2 Evaluation metrics

I report performance using three standard regression metrics:

- Mean Absolute Error (MAE): average absolute deviation between predicted and true scores.
- Root Mean Squared Error (RMSE): square root of the average squared error, which penalises large errors more strongly.
- Coefficient of determination (R^2): proportion of variance explained by the model relative to the mean baseline.

MAE and RMSE are reported in score points. Because the target range is narrow (80–100), even small differences in MAE can be practically meaningful.

4 Results

This section presents the empirical results of the three models under the 3-fold cross-validation protocol described in Section 3. All numbers are averaged over the three folds, and metrics are reported on the test portions of the data unless stated otherwise.

4.1 Overall performance (Table 3)

Table 3 summarises the 3-fold cross-validation performance of the three models on the 10,000-sample subset. The mean baseline attains an MAE of about 2.48 and an RMSE of 3.02, with $R^2 \approx 0.00$, confirming that predicting a constant average score does not explain any variance in the data. Both KNN and Random Forest substantially improve upon this baseline. KNN reduces the MAE to 1.91 and reaches an R^2 of 0.36, while the Random Forest achieves the best performance with an MAE of 1.84, an RMSE of 2.32, and R^2 of 0.41.

Table 3: 3-fold cross-validation performance on the sampled wine subset ($n = 10,000$).

Model	MAE	RMSE	R^2
Baseline (mean)	2.48	3.02	0.00
KNN ($k = 5$)	1.91	2.42	0.36
Random Forest	1.84	2.32	0.41

4.2 Model comparison and overfitting

Figure 4 shows bar plots of the test MAE, and Figure 5 shows the test R^2 for the three models. The baseline has the highest MAE and $R^2 \approx 0$, whereas KNN and Random Forest both yield lower errors and positive R^2 values. The Random Forest consistently outperforms KNN, but the improvement is modest, indicating that price and basic metadata carry limited predictive power.

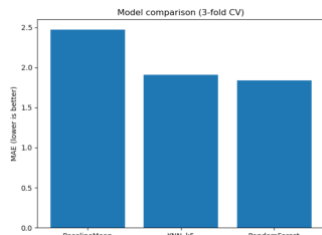


Figure 4: Test MAE comparison across the three models under 3-fold cross-validation (lower is better).

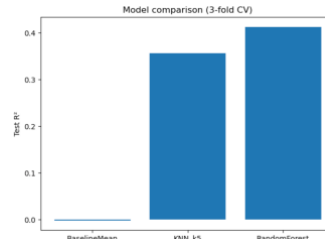


Figure 5: Test R^2 comparison across the three models under 3-fold cross-validation (higher is better).

4.3 Feature importance

Figure 6 reports the grouped feature importances from the Random Forest. The log-transformed price

feature is by far the most important predictor, accounting for roughly 62% of the total importance. Winery contributes about 24%, while variety and regional attributes (province and country) have smaller but non-negligible contributions. This pattern matches domain intuition: more expensive wines from reputable wineries and certain grape varieties tend to receive higher scores, while region plays a secondary role.



Figure 6: Train vs test R^2 for the three models under 3-fold cross-validation.

4.4 Feature importance and model interpretation (Figure 7)

Beyond aggregate error metrics, it is important to understand which features drive the predictions of the best-performing model. For the Random Forest, I compute feature importances and aggregate them by original attribute (price, country, province, variety, winery). The resulting grouped importances are displayed in Figure 7.

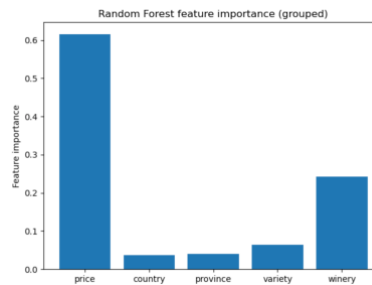


Figure 7: Aggregated Random Forest feature importance for price, country, province, variety, and winery.

Figure 7 shows that the log-transformed price feature is by far the most important predictor, accounting for roughly 62% of the total importance. The winery attribute contributes around 24%, while variety accounts for about 6%. The remaining importance is split between province and country, each contributing roughly 3–4%.

This ranking aligns well with domain intuition: more expensive wines from reputable wineries and certain grape varieties tend to receive higher critic scores, and regional effects are present but weaker. At the same time, the relatively modest R^2 values in Table 3 indicate that a substantial amount of variation remains unexplained, which is plausible given the subjective nature of wine ratings and the fact that many relevant factors (e.g., vintage, detailed tasting notes, winemaking techniques) are not explicitly modelled in this study.

5 Discussion

This project extends the mid-term work in three directions: adding a mean baseline, using k-fold cross-validation, and comparing two non-linear models under a unified preprocessing pipeline. The results show that each of these extensions improves the credibility of the study.

First, the mean baseline provides a clear lower bound. With an MAE of about 2.48 and $R^2 \approx 0$, it confirms that simply predicting the average score is not informative. Both KNN and Random Forest reduce the MAE by roughly half a point and achieve positive R^2 values, which demonstrates that the structured features (price and basic metadata) contain meaningful signal.

Second, replacing a single train–test split with 3-fold cross-validation leads to more stable estimates of generalisation error and directly addresses the mid-term feedback. The small standard deviations of MAE and R^2 across folds indicate that the ranking of models (Random Forest > KNN >> baseline) is

not an artefact of a particular random split.

Third, the comparison between KNN and Random Forest highlights the effect of model capacity. KNN already explains around 30% of the variance, while Random Forest pushes this to about 41% at the cost of mild overfitting, as seen in the train–test R^2 gap. The feature importance analysis provides a plausible interpretation: log-transformed price is the dominant predictor, followed by winery and, to a lesser extent, variety and regional attributes. At the same time, the relatively modest R^2 values suggest that many influential factors, such as vintage or detailed tasting descriptors, are missing from the current feature set.

6 Conclusions and Future Work

This report studied a supervised regression task on the WineEnthusiast dataset: predicting expert wine scores from basic metadata. Using a consistent preprocessing pipeline and 3-fold cross-validation, I compared a mean baseline, a KNN regressor, and a Random Forest regressor on a 10,000-sample subset.

The main findings are that (i) both KNN and Random Forest substantially outperform the mean baseline, confirming that the selected features are informative; (ii) Random Forest achieves the best overall performance, with an MAE of 1.84 and R^2 of 0.41; and (iii) train–test comparisons show only moderate overfitting, indicating that the chosen hyperparameters provide a reasonable bias–variance trade-off. Feature importance analysis further reveals that price and winery are the key drivers of the Random Forest’s predictions.

Future work may extend this analysis in several directions. First, rather than relying solely on randomly shuffled cross-validation, models could be evaluated under explicitly time-ordered train–test splits, training on earlier vintages or review years and testing on later ones, in order to more closely approximate real-world deployment where predictions are required for future wines. Second, model performance could be examined across well-defined subgroups such as country of origin, production region and price segment, for example distinguishing between budget, mid-range and premium wines, to identify systematic performance differences and potential sources of bias and to motivate segment-specific modelling strategies where appropriate. Third, the Random Forest framework could be extended to provide formal uncertainty quantification, for instance by employing quantile regression forests or related approaches, so that the system outputs both point predictions and associated prediction intervals, thereby offering more informative support for downstream tasks such as inventory management and price optimization.

7 Reference

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- [2] F. Pedregosa et al. “Scikit-learn: Machine Learning in Python.” *Journal of Machine Learning Research*, 12, 2011.
- [3] L. Breiman. “Random Forests.” *Machine Learning*, 45(1), 2001.

8 Author Contributions

Renna Wu and Xiaoxin Lao contributed equally to this project. Together, they selected the Wine Reviews dataset, defined the research objectives, and designed the preprocessing pipeline and modelling approach (baseline, KNN, and Random Forest with cross-validation). Both authors jointly carried out data preprocessing, model implementation, experiments, and result analysis, and they co-wrote and revised the final manuscript. In addition, Renna Wu prepared the presentation slides, and Xiaoxin Lao was responsible for the video recording. The order of authors reflects equal contribution.